

1 Solution — GIAM 2.1

Problem 3

1.1 Problem

Consider the sentence “This sentence is false.” Is this sentence a statement?

1.2 Answer

No — it is **not** a statement.

This is the famous **liar paradox**. A statement must have a *definite, consistent* truth value — true or false. The liar sentence can be assigned *neither* without contradiction, so it fails the very definition of a statement.

1.3 The Reasoning

Call the sentence L . There are only two possibilities to test — L true or L false — and both self-destruct.

Try each truth value in turn. Both assumptions collapse into their opposite — an endless flip.

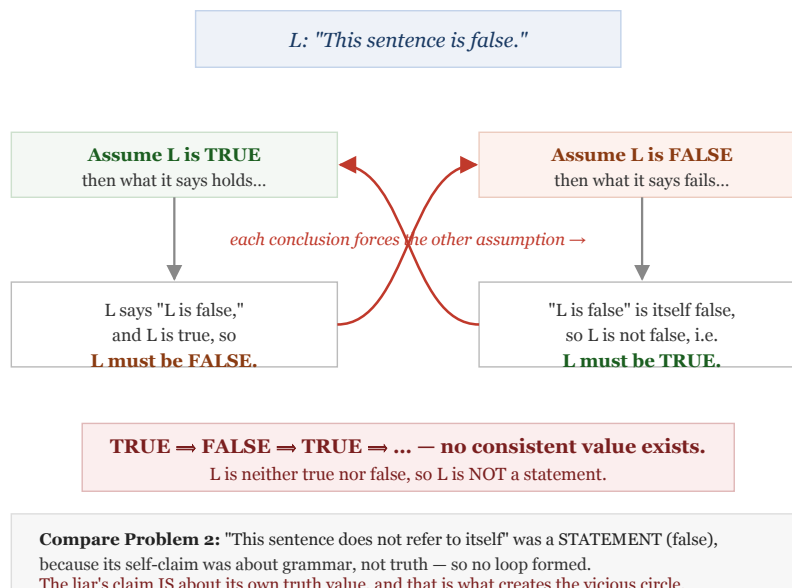


Figure 1.1: An analysis of $L = \text{'This sentence is false.'}$ Left branch: assume L is TRUE; then what it says holds, and it says ' L is false', so L must be FALSE. Right branch: assume L is FALSE; then ' L is false' is itself false, so L is not false, i.e. L must be TRUE. Red arrows show each conclusion forcing the opposite assumption: TRUE gives FALSE gives TRUE forever, with no consistent value. So L is neither true nor false and is NOT a statement. A contrast box notes Problem 2's sentence WAS a statement (false) because its self-claim was about grammar not truth, so no loop formed; the liar's claim is about its own truth value, which creates the vicious circle.

Case 1 — suppose L is true. Then what L asserts is the case. But L asserts " L is false." So L is false — contradicting our supposition.

Case 2 — suppose L is false. Then what L asserts fails to hold. L asserts " L is false," so *that* is false, meaning L is *not* false — i.e. L is true — again contradicting the supposition.

Each assumption forces its own opposite:

$$L \text{ true} \implies L \text{ false} \implies L \text{ true} \implies \dots$$

an unending flip with no stable resting point. Since no consistent truth value can be attached, L is **not a statement**.

1.4 Remark — The Sharp Contrast With Problem 2

This problem is the deliberate foil to the previous one. There, "This sentence does not refer to itself" turned out to *be* a statement (a false one); here, "This sentence is false" is *not* a statement at all. The single decisive difference is **what the self-reference is about**:

	Problem 2 sentence	Problem 3 (liar)
self-claim is about...	a <i>grammatical</i> property (does it refer to itself?)	its <i>own truth value</i>
answer fixed independently of truth?	yes (it plainly does refer to itself)	no (truth value is the very unknown)

result	consistently false	— a statement	no consistent value	— not a statement
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Table 1.1

When a sentence talks about a feature settled *before* truth enters (word count, language, reference), self-reference is harmless. When it talks about its *own* truth, the evaluation feeds back into itself and the vicious circle appears.

1.5 Remark — Why This Does Not Break Logic

A first reaction is alarm: has logic sprung a leak? No. The resolution is simply to **restrict the domain**. Classical logic is built only for sentences that *are* statements — those with a definite truth value — and the liar is excluded from that domain at the outset, exactly as questions and commands are. The liar does not prove a *contradiction within* logic (which would be catastrophic, since from a contradiction everything follows); it merely exhibits a sentence that lies *outside* logic’s intended scope. Recognising and quarantining such sentences is precisely why the chapter is so careful to define “statement” before doing anything else.

1.6 Remark — Escaping the Paradox: Tarski’s Hierarchy

Logicians have made the informal fix precise. Alfred Tarski’s diagnosis is that a consistent language cannot contain its own truth predicate — “is true”/“is false” for sentences of language $\mathcal{L}$ must live in a richer *meta-language* $\mathcal{L}'$. In that stratified setup the liar cannot even be *written*, because it would have to apply a level- n truth predicate to a level- n sentence, which the hierarchy forbids. Other approaches keep one language but change the logic — e.g. Kripke’s *truth-value gaps*, where the liar is simply assigned *neither* true nor false (it is “ungrounded”). Both routes agree with our elementary conclusion: the liar is not a legitimate bearer of a truth value.

1.7 Remark — The Same Engine Powers Deep Theorems

The liar looks like a mere curiosity, but the self-referential machinery behind it is one of the most productive ideas in all of mathematics — *when aimed correctly*. Replace “false” with “not provable” and you get Gödel’s sentence “this statement is not provable,” which (unlike the liar) is **consistent** and yields the **Incompleteness Theorems**: any sufficiently strong formal system contains true statements it cannot prove. Replace it with “does not halt” applied to programs and you get Turing’s proof that the **Halting Problem** is undecidable. Russell’s paradox — “the set of all sets that do not contain themselves” — is the set-theoretic liar, and forced the axiomatic rebuilding of set theory. The liar sentence of this exercise is thus the toy model for a family of results that map the outer limits of computation, provability, and set theory.